

## YERBOGACHEN, Russian Federation

WMO: 248170

Lat: 61.276N

Lon: 108.032E

Elev: 291

StdP: 97.88

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -48.0      | -45.4 | -49.9                         | 0.0 | -47.8 | -47.3 | 0.0 | -45.2 | 6.3                   | -19.9 | 5.5 | -19.4 | 0.1                   | 50   | 0.692 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 29.3            | 18.1 | 27.2 | 17.2 | 25.1 | 16.3 | 19.3                | 26.3 | 18.2 | 25.0 | 17.3 | 23.5 | 2.4                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.0                            | 12.5 | 21.6 | 15.7 | 11.6 | 20.5 | 14.8 | 10.9 | 19.7 | 56.4          | 26.6 | 53.0 | 25.0 | 49.9 | 23.8 | 24.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 6.2               | 5.3  | 4.5 | -49.8                      | 32.8 | 2.4                | 1.3 | -51.5                                              | 33.7 | -52.9      | 34.5 | -54.2      | 35.2 | -55.9      | 36.2 |    |    |
|                   |      |     | -49.7                      | 21.2 | 2.3                | 1.3 | -51.4                                              | 22.1 | -52.8      | 22.8 | -54.1      | 23.5 | -55.8      | 24.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -5.8   | -29.2 | -25.9 | -14.5 | -3.4  | 5.6   | 15.0  | 17.1  | 13.5  | 5.6   | -4.7  | -20.9 | -28.9 |
|                                                                                  | DBStd       | 18.21  | 9.31  | 8.43  | 8.58  | 5.92  | 4.42  | 4.19  | 3.52  | 3.68  | 4.38  | 6.51  | 10.10 | 10.33 |
|                                                                                  | HDD10.0     | 6276   | 1215  | 1005  | 759   | 402   | 148   | 7     | 1     | 12    | 142   | 455   | 926   | 1205  |
|                                                                                  | HDD18.3     | 8843   | 1473  | 1238  | 1017  | 652   | 394   | 114   | 66    | 155   | 382   | 713   | 1176  | 1463  |
|                                                                                  | CDD10.0     | 521    | 0     | 0     | 0     | 0     | 12    | 158   | 220   | 119   | 11    | 0     | 0     | 0     |
|                                                                                  | CDD18.3     | 47     | 0     | 0     | 0     | 0     | 0     | 16    | 27    | 4     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 841    | 0     | 0     | 0     | 0     | 3     | 318   | 409   | 108   | 2     | 0     | 0     | 0     |
|                                                                                  | CDH26.7     | 232    | 0     | 0     | 0     | 0     | 0     | 94    | 112   | 26    | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 1.8    | 1.3   | 1.4   | 1.9   | 2.2   | 2.4   | 2.0   | 1.9   | 1.8   | 1.9   | 2.0   | 1.5   | 1.2   |
| Precipitation                                                                    | PrecAvg     | 338    | 16    | 11    | 12    | 17    | 29    | 46    | 46    | 51    | 35    | 33    | 26    | 19    |
|                                                                                  | PrecMax     | 475    | 39    | 30    | 32    | 34    | 61    | 127   | 120   | 139   | 77    | 83    | 59    | 42    |
|                                                                                  | PrecMin     | 173    | 3     | 1     | 2     | 4     | 5     | 3     | 4     | 6     | 6     | 10    | 9     | 4     |
|                                                                                  | PrecStd     | 65     | 7     | 6     | 7     | 7     | 14    | 23    | 25    | 32    | 18    | 16    | 10    | 9     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | -6.9  | -2.7  | 7.7   | 13.6  | 23.1  | 32.1  | 32.1  | 29.8  | 22.5  | 10.3  | 1.6   | -3.2  |
|                                                                                  |             | MCWB   | -7.5  | -3.8  | 1.9   | 6.0   | 12.6  | 18.2  | 19.4  | 18.9  | 13.4  | 6.4   | 0.4   | -3.8  |
|                                                                                  | 2%          | DB     | -9.9  | -5.8  | 3.5   | 10.8  | 19.9  | 29.4  | 29.5  | 26.2  | 18.4  | 7.5   | -0.1  | -6.0  |
|                                                                                  |             | MCWB   | -10.5 | -6.9  | -0.5  | 4.3   | 10.0  | 17.6  | 18.5  | 17.0  | 11.4  | 4.7   | -1.1  | -6.4  |
|                                                                                  | 5%          | DB     | -12.5 | -9.1  | 0.6   | 8.5   | 17.7  | 26.9  | 27.6  | 23.7  | 15.6  | 5.1   | -2.0  | -9.8  |
|                                                                                  |             | MCWB   | -13.0 | -10.0 | -2.1  | 2.8   | 8.9   | 16.3  | 17.9  | 15.7  | 10.4  | 2.9   | -2.9  | -10.2 |
|                                                                                  | 10%         | DB     | -15.1 | -12.4 | -1.5  | 6.0   | 15.0  | 24.6  | 25.6  | 21.5  | 13.0  | 3.0   | -5.2  | -13.2 |
|                                                                                  |             | MCWB   | -15.4 | -13.0 | -3.8  | 1.5   | 7.5   | 15.5  | 17.1  | 14.9  | 9.2   | 1.5   | -5.9  | -13.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -7.4  | -3.8  | 2.5   | 6.6   | 13.1  | 19.9  | 21.1  | 19.4  | 14.5  | 6.9   | 0.7   | -3.6  |
|                                                                                  |             | MCDB   | -6.9  | -2.9  | 7.0   | 12.0  | 21.0  | 27.1  | 28.7  | 27.3  | 19.6  | 9.6   | 1.3   | -3.1  |
|                                                                                  | 2%          | WB     | -10.5 | -6.8  | -0.1  | 4.7   | 11.0  | 18.5  | 19.9  | 17.8  | 12.5  | 5.0   | -1.0  | -6.4  |
|                                                                                  |             | MCDB   | -10.1 | -5.9  | 3.0   | 9.8   | 17.4  | 26.5  | 26.8  | 24.2  | 16.8  | 7.4   | -0.1  | -5.8  |
|                                                                                  | 5%          | WB     | -12.9 | -10.0 | -2.0  | 3.3   | 9.4   | 17.5  | 18.9  | 16.7  | 11.0  | 3.3   | -3.0  | -10.3 |
|                                                                                  |             | MCDB   | -12.6 | -9.2  | 0.5   | 7.8   | 16.3  | 24.9  | 25.2  | 22.1  | 14.9  | 4.7   | -2.0  | -9.9  |
|                                                                                  | 10%         | WB     | -15.5 | -13.0 | -3.7  | 1.9   | 8.0   | 16.5  | 17.9  | 15.6  | 9.5   | 1.6   | -5.9  | -13.5 |
|                                                                                  |             | MCDB   | -15.1 | -12.5 | -1.5  | 5.5   | 14.4  | 23.0  | 24.0  | 20.3  | 12.7  | 2.9   | -5.2  | -13.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.5   | 13.5  | 16.1  | 12.5  | 11.0  | 13.0  | 12.3  | 12.1  | 9.9   | 7.5   | 9.9   | 8.9   |
|                                                                                  |             | MCDBR  | 9.6   | 11.9  | 14.1  | 14.2  | 16.9  | 18.0  | 17.0  | 17.1  | 15.2  | 9.4   | 8.9   | 8.7   |
|                                                                                  |             | MCWBR  | 9.3   | 11.1  | 11.1  | 9.0   | 8.9   | 7.7   | 7.5   | 8.6   | 8.9   | 6.8   | 8.5   | 8.6   |
|                                                                                  | 5% WB       | MCDBR  | 9.5   | 11.9  | 13.6  | 12.0  | 14.6  | 15.2  | 14.2  | 14.1  | 13.2  | 8.1   | 9.1   | 8.7   |
|                                                                                  |             | MCWBR  | 9.2   | 11.0  | 11.0  | 7.9   | 8.5   | 7.1   | 7.1   | 7.7   | 8.5   | 6.3   | 8.7   | 8.5   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.238  | 0.279 | 0.306 | 0.336 | 0.364 | 0.394 | 0.456 | 0.421 | 0.332 | 0.293 | 0.235 | 0.205 |       |
|                                                                                  | taud        | 1.945  | 2.016 | 2.052 | 2.103 | 2.295 | 2.332 | 2.133 | 2.196 | 2.445 | 2.287 | 1.980 | 1.874 |       |
|                                                                                  | Ebn at Noon | 446    | 669   | 794   | 847   | 856   | 832   | 759   | 754   | 784   | 671   | 444   | 306   |       |
|                                                                                  | Edh at Noon | 44     | 79    | 109   | 127   | 115   | 113   | 135   | 117   | 76    | 60    | 40    | 27    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.40   | 1.35  | 3.00  | 4.81  | 5.27  | 5.79  | 5.23  | 4.00  | 2.43  | 1.09  | 0.52  | 0.21  |       |
|                                                                                  | RadStd      | 0.02   | 0.09  | 0.17  | 0.30  | 0.34  | 0.44  | 0.37  | 0.31  | 0.21  | 0.13  | 0.05  | 0.01  |       |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | N/A   | N/A     | N/A    | +2.29   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | +2.29   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page